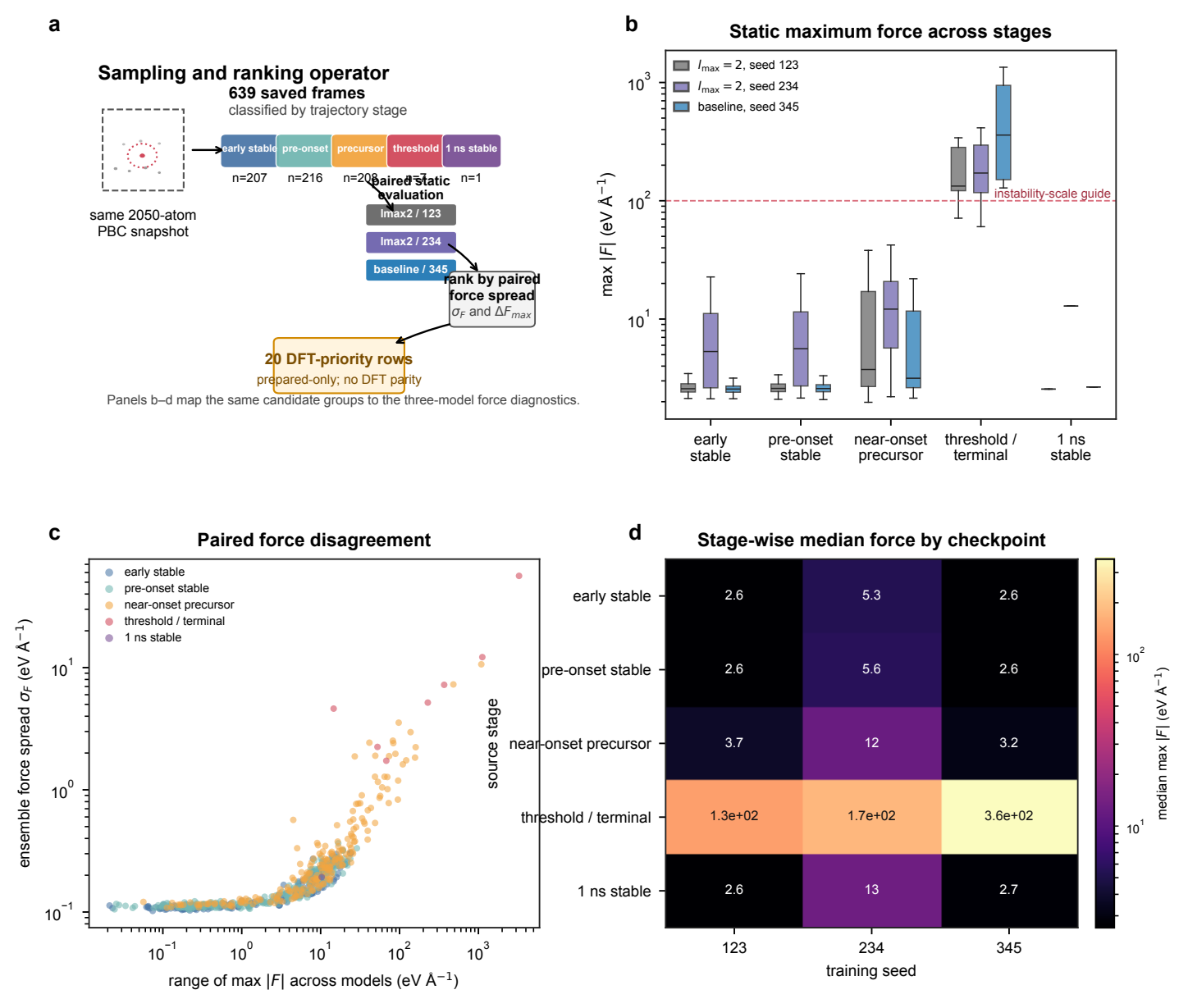

a
Sampling and ranking operator
639 saved frames
classified by trajectory stage
same 2050-atom
PBC snapshot
early stable
pre-onset
precursor
threshold
1 ns stable
n=207
n=216
n=208
n=7
n=1
paired static evaluation
lmax2 / 123
lmax2 / 234
baseline / 345
rank by paired force spread
σ_F and ΔF_{max}
20 DFT-priority rows
prepared-only; no DFT parity
Panels b–d map the same candidate groups to the three-model force diagnostics.
b
Static maximum force across stages
l_{max} = 2, seed 123
l_{max} = 2, seed 234
baseline, seed 345
instability-scale guide
max |F| (eV Å^{-1})
early stable
pre-onset stable
near-onset precursor
threshold / terminal
1 ns stable
c
Paired force disagreement
early stable
pre-onset stable
near-onset precursor
threshold / terminal
1 ns stable
source stage
ensemble force spread σ_F (eV Å^{-1})
range of max |F| across models (eV Å^{-1})
d
Stage-wise median force by checkpoint
early stable
pre-onset stable
near-onset precursor
threshold / terminal
1 ns stable
2.6
5.3
2.6
2.6
5.6
2.6
3.7
12
3.2
1.3e+02
1.7e+02
3.6e+02
2.6
13
2.7
123
234
345
training seed
median max |F| (eV Å^{-1})